



Development of Photon Readout Electronics Prototypes for nEXO

Sirray Smith, Department of Physics and Engineering Physics, Morgan State University, Baltimore, MD 21251
Dr. Stacyann Nelson, Department of Physics and Engineering Physics, Morgan State University, Baltimore, MD 21251
Dr. Mickey Chiu, Physics Department, Brookhaven National Laboratory, Upton NY 11973

Abstract

nEXO is a 5-ton liquid Xenon time projection chamber (TPC) operating at cryogenic temperatures that will search for neutrino-less double beta decay ($0\nu\beta\beta$). In this project we will contribute to improving the nEXO photon readout by evaluating the performance of prototypes that are designed to be more radiopure by having fewer capacitors. To achieve this, we used Python to control the electronics, and to analyze data gathered from the prototypes. The LArASIC (Liquid Argon Application Specific Integrated Circuit) chip is being used to read out the photon signals from nEXO and here at this early stage of prototyping, we are using a modified LArASIC board.

Importance of nEXO and LArASIC

- A neutrino is the lightest mass particle in the standard model. Formerly its mass was believed to be zero.
- nEXO may answer the question if a neutrino is a Majorana particle. i.e., if the neutrino is its own antiparticle.
- nEXO may explain some of the deepest unanswered questions about particle physics: how neutrinos have mass, discover lepton number violation, and explain how the universe become matter dominated.
- The LArASIC is being used to read out the photon signals from nEXO.
- nEXO will have sensitivity to a half-life of 10^{-28} years, which would be the rarest radioactive decay ever.
- This incredibly difficult measurement can only be done by reducing the radioactive background to miniscule levels.
- Our studies will determine the minimum number of capacitors required by the LArASIC while keeping within the requirements of noise and cross-talk.

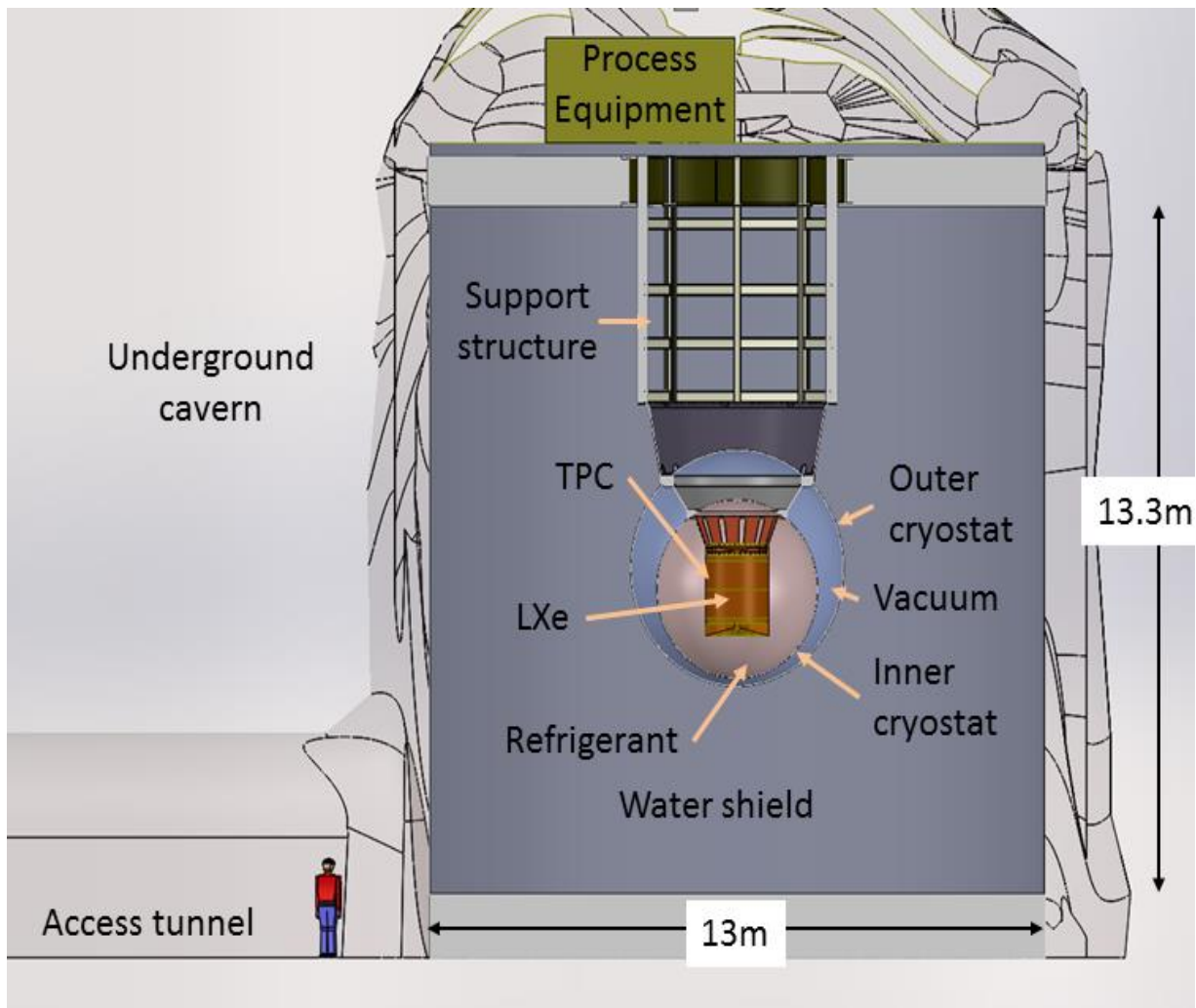


Figure 1: Illustration of nEXO

The LArASIC Chip

The LArASIC board, shown in Figure 2, is used for determining the noise and cross-talk at room-temperature. These are then measured again with the board submerged in liquid nitrogen. The currents on the power supply for the 1.8V should read 0.005 Amps and the current for the power supply for the 5V should read 1.04 Amps, when turned on. The data acquisition computer is connected to the LArASIC board through ethernet, and python code is executed to collect the noise data. The noise analysis code is then run afterwards. The process repeats for collecting the cross-talk using the xtalk code. The power supplies are turned off. The dewar is then filled with liquid nitrogen and the LArASIC board is placed in. The power supplies are turned back on. The same code is run again for both the noise and the cross-talk.

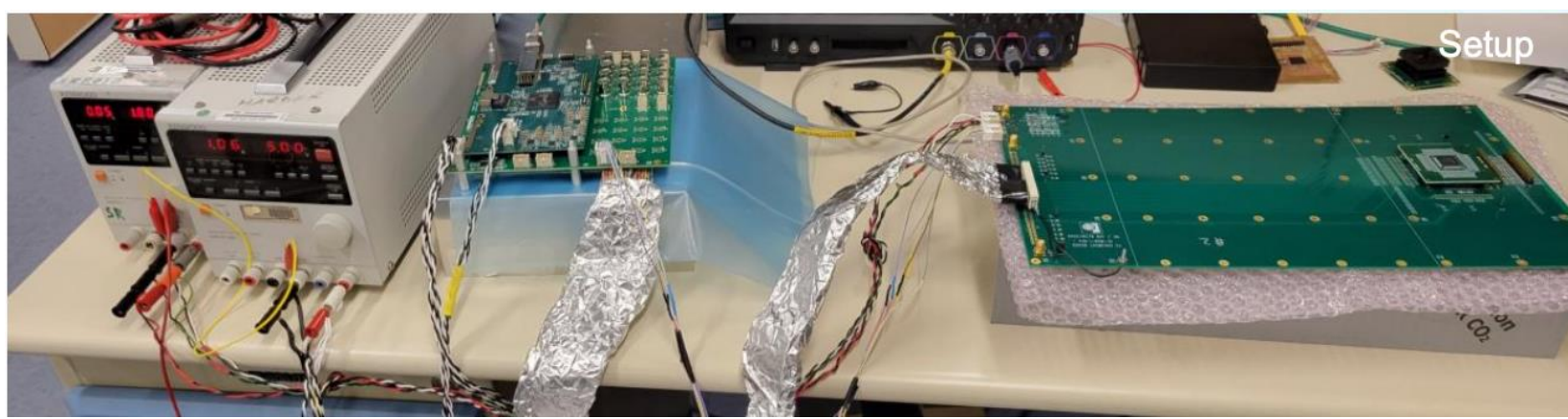


Figure 2: The setup for the LArASIC.

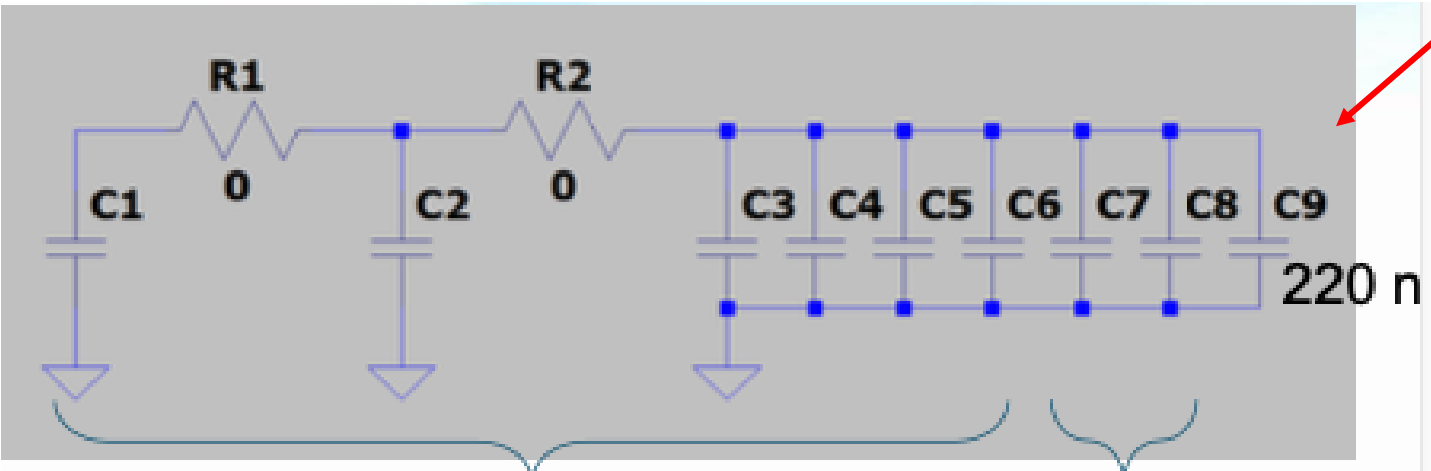


Figure 3: The LArASIC board.

Results

Noise

Figure 4 and Figure 5 show the comparison of the noise for all the channels while the board is room-temperature. Figure 4 is the noise before any capacitors are removed. Figure 5 is the noise with only one capacitor of 1 x200 uF on the VDDP.

The noise is measured by taking the average root-mean-square from the numerous channels and samples, after being collected and transformed from the raw waveform data. The measurements are shown as the Equivalent Noise in terms of Charge (ENC). The capacitors on the VDDO did not seem have to any affect on the ENC levels. They seemed to stay relatively the same throughout testing. The capacitors on the VDDP seemed to have a major affect on the ENC levels. Removing majority of the capacitors on the VDDP seemed to cause the ENC levels to almost double for the longer gain and shaping times.

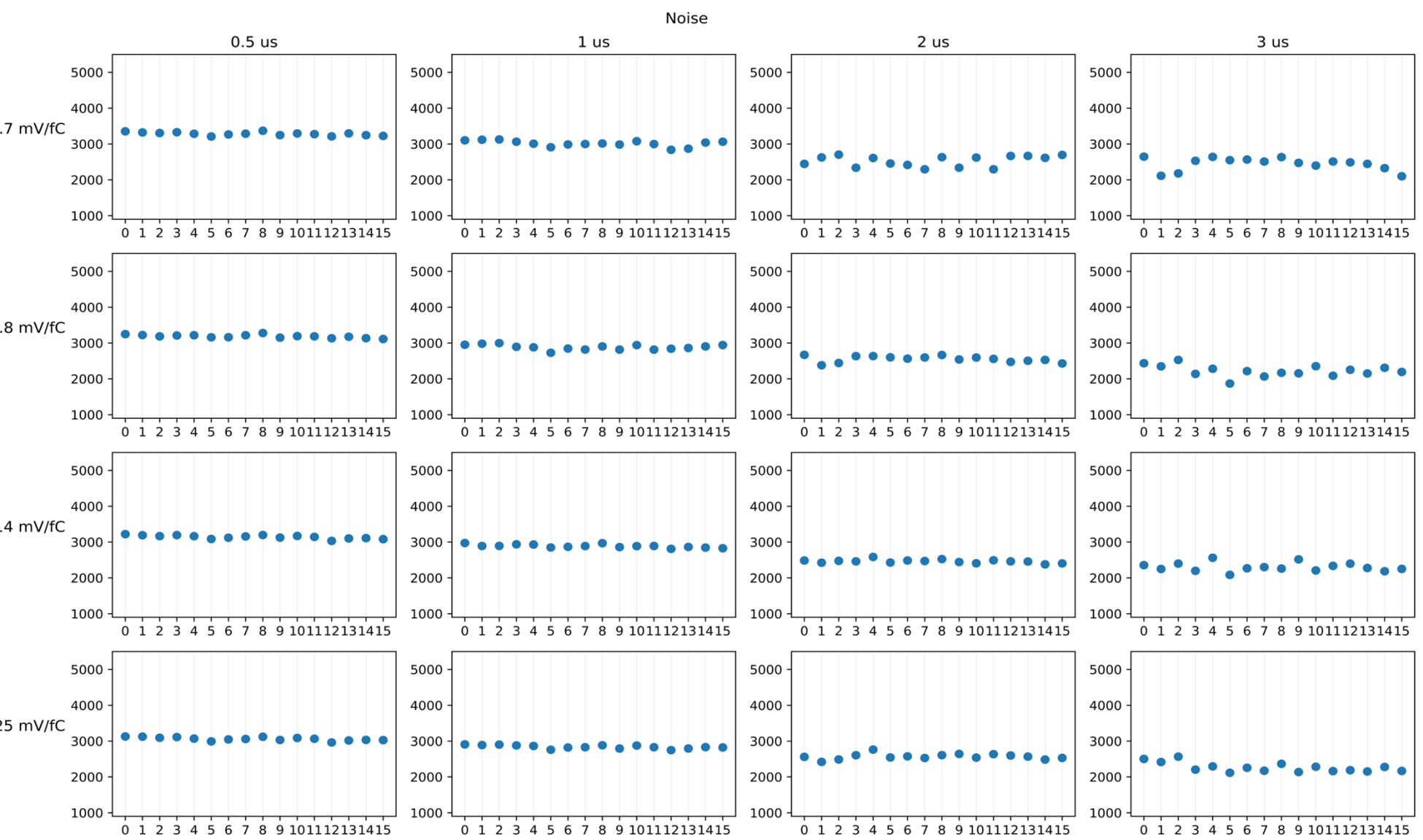


Figure 4: Noise plots, All Capacitors

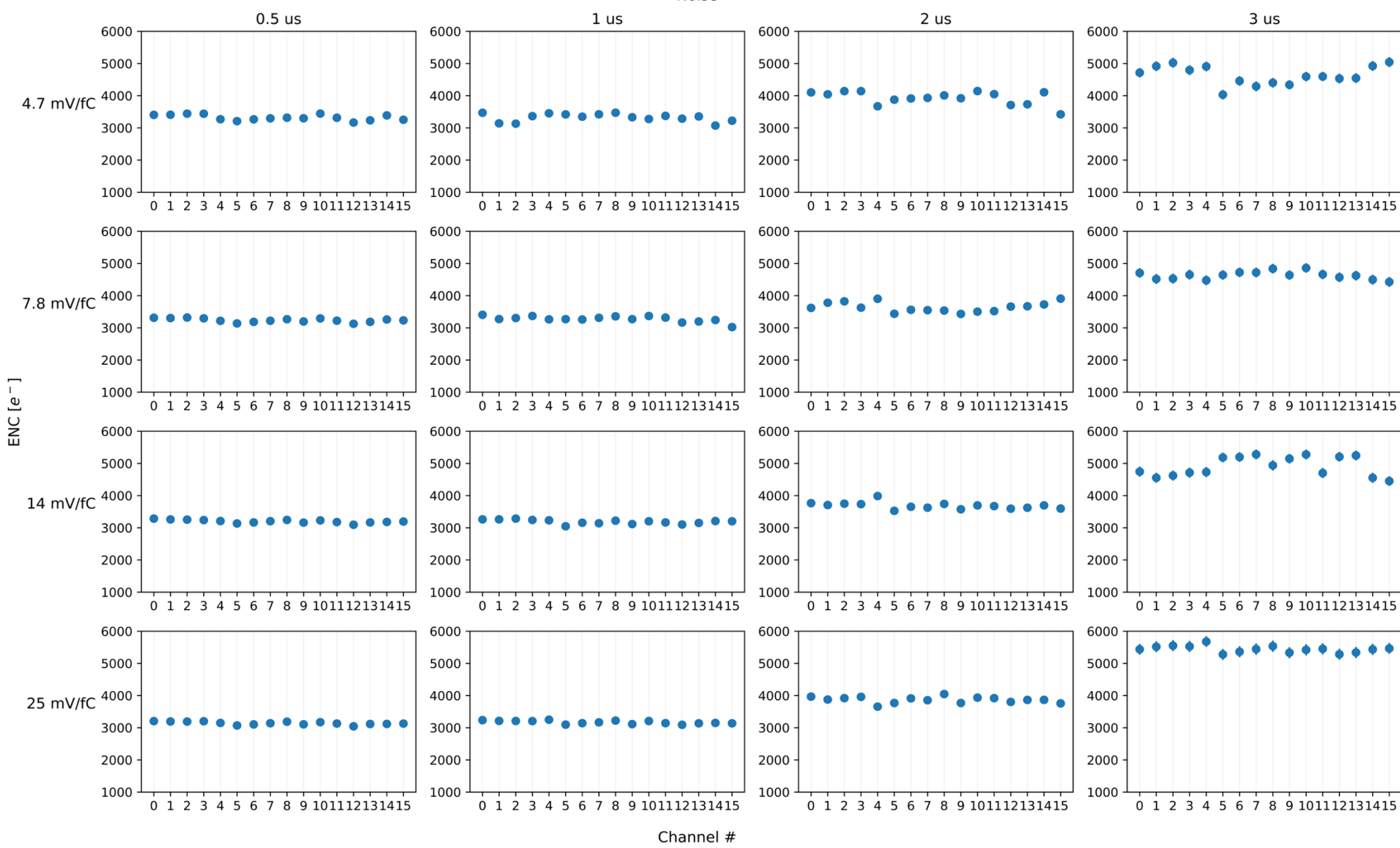


Figure 5: Noise plots, Only One Capacitor, VDDP 1x200 uF

Capacitor Configuration

PS	Warm	Cold
VDD	6x220 uF 2x10 uF 1x220 nF	14x10 uF 7x220 nF
VDDP	6x220 uF 2x10 uF 1x220 nF	8x10 uF 4x 220 nF
VDDO	6x220 uF 2x10 uF 1x220 nF	4x10 uF 2x 220 nF

PS	Warm	Cold
VDD	0x220 uF 0x10 uF 0x220 nF	2x10 uF 7x220 nF
VDDP	1x220 uF 0x10 uF 0x220 nF	8x10 uF 4x 220 nF
VDDO	0x220 uF 0x10 uF 0x220 nF	1x10 uF 2x 220 nF

Crosstalk

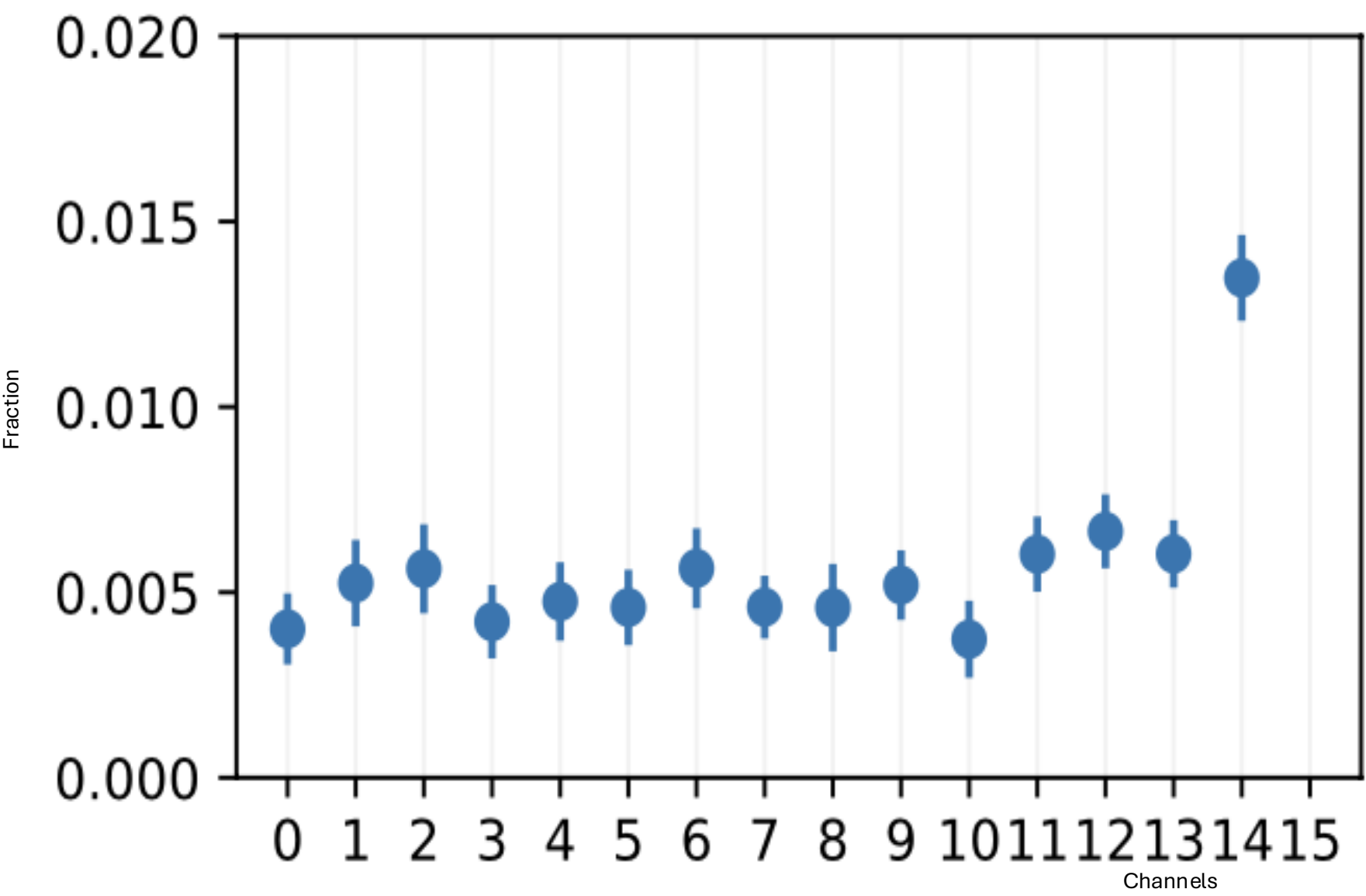


Figure 6: Channel 15 Crosstalk, with gain 25 mV/FC, All capacitors

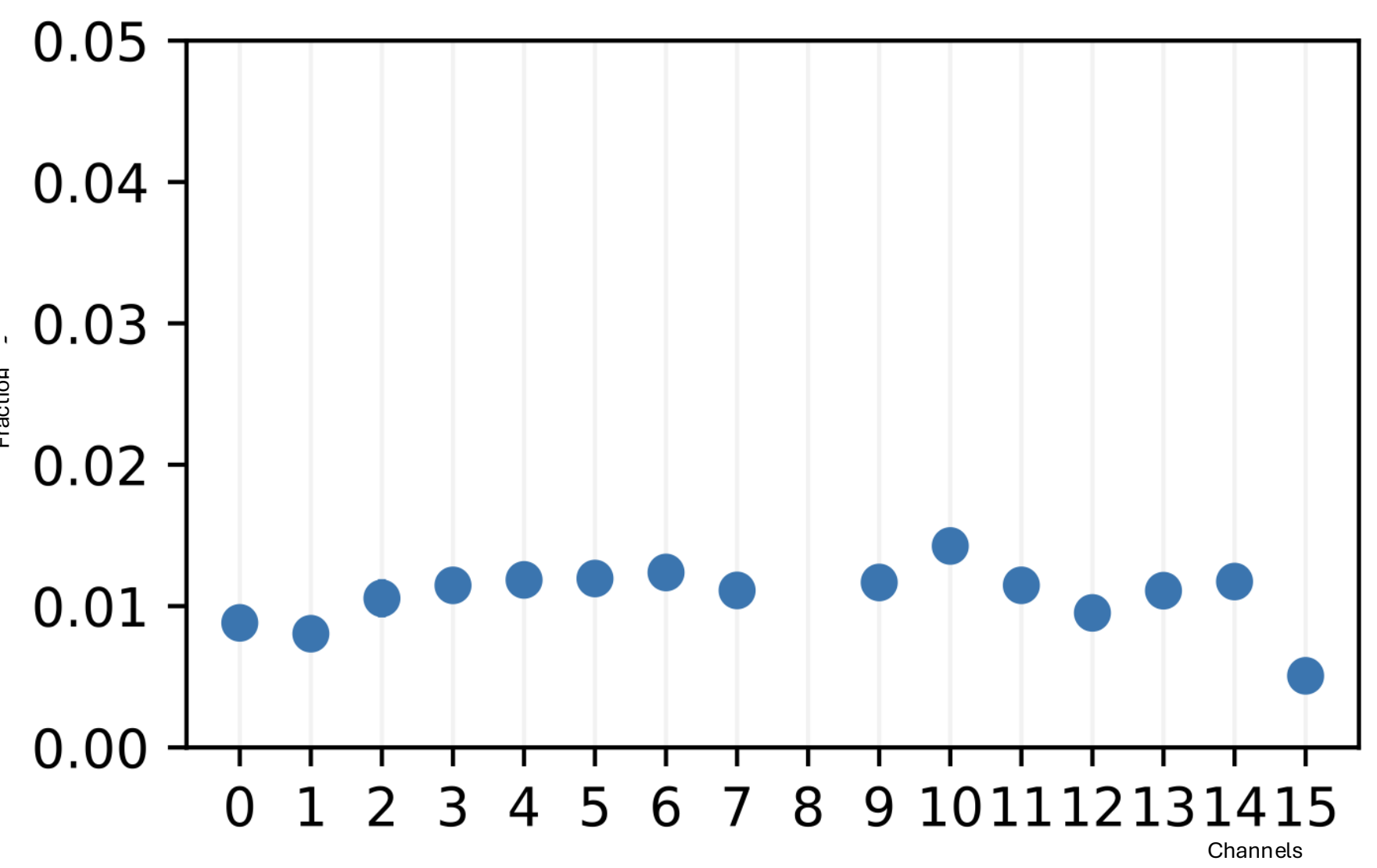


Figure 7: Channel 15 Crosstalk, with gain 25 mV/FC, One capacitor on VDDP 1 x 200 uF

Figure 6 and Figure 7 show the comparison of crosstalk for Channel 8 when the board is warm room-temperature. Figure 6 is the cross-talk before any capacitors are removed. Figure 7 is the cross-talk after capacitors are removed. The other channel's crosstalk looks very similar to Channel 8. The crosstalk was only tested for 25 mV/FC at 0.5 us for this run.

The crosstalk is measured by sending a pulse to one channel and reading out all the channels to see if there is a response from any of the other channels. The amplitude is read for all the channels, and the ratio is taken with the amplitude of the injected pulse. The ratio is the fractional crosstalk. Ideally, the goal is from there to be zero crosstalk from other channels.

Conclusion

The LArASIC for nEXO application is still undergoing testing. The noise levels seem to increase as the number of capacitors lessen. The noise seems to increase by a factor of 2. The cross-talk, also, seems to increase as the number of capacitors lessen. The measurements for cross-talk for all four gain and shaping times while the board is room-temperature is still ongoing. The measurements for noise and cross-talk while the board is submerged in liquid nitrogen is still ongoing. Overall, the LArASIC for nEXO is still under development and work is being done to lower the noise and cross-talk levels.

Acknowledgments

This material is based upon work supported by the U.S. Department of Energy, Office of Science, Office of Nuclear Physics, Reaching a New Energy Sciences Workforce (NP-RENEW). I would also like to thank Dr. Chiu and Dr. Gallice, as well as Professor Nelson, for their mentorship. No export control.

References

1. Agostini, M., Benato, G., & Detwiler, J. A. (2017, September 7). Discovery probability of next-generation neutrinoless double-beta decay experiments. *Physical Review D*. <https://doi.org/10.1103/PhysRevD.96.053001>
2. C. Grace et al., "ColdADC: P2: A 16-Channel Cryogenic ADC ASIC for the Deep Underground Neutrino Experiment," in *IEEE Transactions on Nuclear Science*, vol. 69, no. 1, pp. 105-112, Jan. 2022, doi: 10.1109/TNS.2021.3136404.
3. XENON Collaboration*, Observation of two-neutrino double electron capture in ^{124}Xe with XENON1T. *Nature* **568**, 532–535 (2019). <https://doi.org/10.1038/s41586-019-1124-4>

